

Indian Institute of Engineering Science and Technology, Shibpur
Dual Degree (B.Tech-M.Tech) 3rd Semester Final Examination
November 2016

Subject: Mathematics-III (MA-301)

Time : 3 hours

Full Marks : 70

(Use separate answer script for each half)

First Half (Full Marks-40)

Answer any eight questions

1. State axiomatic definition of probability. Show that for any two events A and B :

$$P(\overline{A}B) = P(B) - P(AB).$$

[5]

2. State and prove Baye's theorem.

[5]

3. The chance that a doctor will diagnose a certain disease correctly is 60%. The chance that a patient will die by his treatment after correct diagnosis is 40% and the chance of death by wrong diagnosis is 70%. A patient of the doctor, who had the disease, dies. What is the probability that the disease was diagnosed correctly?

[5]

4. Find the mean and variance of the Binomial distribution.

[5]

5. Suppose that an airplane engine will fail, when in flight, with probability $(1 - p)$ independently from engine to engine. Suppose that the airplane will make a successful flight if at least 50% of its engines remain operative. For what values of p is a four-engine plane preferable to a two-engine plane?

[5]

6. Prove that Poisson distribution is a limiting case of Binomial distribution.

[5]

7. State and prove Tchebycheff's inequality for a continuous random variable.

[5]

8. Using Tchebycheff's inequality, find a lower bound for the probability of getting 64 to 184 driving licences issued by Road Transport Authority in a specific month. It is given that the number of driving licences issued per month be a random variable having mean $m = 124$ and standard deviation $\sigma = 7.5$.

[5]

9. State Law of Large Numbers. Suppose X_i takes the values $-\sqrt{2i-1}$ and $\sqrt{2i-1}$ with equal probabilities. Show that the law of large numbers can not be applied to the independent random variables $X_1, X_2, \dots$.

[5]

10. The probability density function of a two-dimensional random variable (X, Y) is given by

$$f(x, y) = \begin{cases} k(x + y), & 0 < x + y < 1 \\ 0, & \text{elsewhere.} \end{cases}$$

Find k and evaluate $P(X < \frac{1}{2}, Y > \frac{1}{4})$.

[5]

11. Prove that the sample mean $\bar{X}$ is an unbiased estimator of the population mean. A random sample $\{X_1, X_2, X_3, X_4\}$ of size 4 is drawn from an infinite population having mean μ and variance σ^2 . Define the two estimators of μ as follows:

$$T_1 = \frac{1}{5}(X_1 + X_2 + X_3 + 2X_4) \text{ and } T_2 = \frac{1}{8}(X_1 + 2X_2 + 3X_3 + 2X_4).$$

Which one is better? Why? Which one is best among $T_1, T_2, \bar{X}$ as an estimator of μ ?

[5]

12. Let $x_1, x_2, \dots, x_n$ be a particular sample of size n drawn from a population with probability density function:

$$f(x, \theta) = 1; \theta - \frac{1}{2} \leq x \leq \theta + \frac{1}{2}, -\infty < \theta < \infty.$$

Obtain maximum likelihood estimate of θ .

[5]

13. Fit a second degree parabola to the following data:

x_i	y_i
0	1
1	1.8
2	1.3
3	2.5
4	6.3

[5]

14. In a partially destroyed Laboratory record of an analysis of correlation data, the following results only are legible:

$$\text{Var}(X) = 9.$$

$$\text{Regression equations: } 8X - 10Y + 66 = 0, 40X - 18Y = 214.$$

Find (i) the mean values of X and Y ,

(ii) the correlation coefficient between X and Y , and

(iii) the standard deviation of Y .

[5]

Answer any THREE questions

15. a) Define Laplace transform of a function $F(t)$. Find the Laplace transform of t^n , when n is a positive but not necessarily an integer.

b) Evaluate (i) $L^{-1} \left\{ \frac{8s+29}{s^2-12s+3^2} \right\}$

(ii) $L^{-1} \left\{ \frac{4}{s^2} + \frac{(\sqrt{s}-1)^2}{s^2} - \frac{5}{3s+4} \right\}$

(1+3) + (3+3)

16. a) If $L\{F(t)\} = f(s)$, then prove that $L\{e^{at} F(t)\} = f(s-a)$.

Hence evaluate $L\{e^{-t}(3 \sin 2t - 5 \cos 2t)\}$.

b) Use convolution theorem to find $L^{-1} \left\{ \frac{s}{(s^2+a^2)^2} \right\}$

(3+2)+5

17. a) Use Laplace transform to solve the differential equation $\frac{d^2 y}{dt^2} + y = 6 \cos 2t$,

given that $y = 3, \frac{dy}{dt} = 1$ when $t = 0$.

- b) Obtain the Laplace inversion of $f(s) = \frac{1}{\sqrt{1+s^2}}$ as a series of powers of t .

6+4

18. a) Explain basic solution of a system of linear equations.

Find all the basic solutions of the set of equations

$$4x_1 + 2x_2 + 3x_3 - 8x_4 = 6$$

$$3x_1 + 5x_2 + 4x_3 - 6x_4 = 8$$

- b) Let $x_1 = 1, x_2 = 3, x_3 = 2$ be a feasible solution of the set of equations

$$2x_1 + 4x_2 - 2x_3 = 10$$

$$10x_1 + 3x_2 + 7x_3 = 33.$$

Reduce the feasible solution to a basic feasible solution.

(2+3)+5

19. a) Define hyperplane and convex set.

Prove that a hyperplane is a convex set.

b) Use simplex method to maximize $5x_1 + 2x_2 + 2x_3$

subject to $2x_1 + 2x_2 - 2x_3 \leq 30$

$$x_1 + 3x_2 + x_3 \leq 36$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

(2+2)+6

20. a) Solve the following L. P. P. by big M method

$$\text{Max} Z = -4x_1 - 3x_2$$

subject to

$$x_1 + 2x_2 \geq 8$$

$$3x_1 + 2x_2 \geq 12$$

$$x_1 \geq 0, x_2 \geq 0$$

b) Solve the following L. P. P. by using two-phase simplex method:

$$\text{Min } Z = x_1 + x_2$$

subject to

$$2x_1 + 4x_2 \geq 4$$

$$x_1 + 7x_2 \geq 7$$

$$x_1 \geq 0, x_2 \geq 0$$

5+5